

Dual Differential Line Receivers

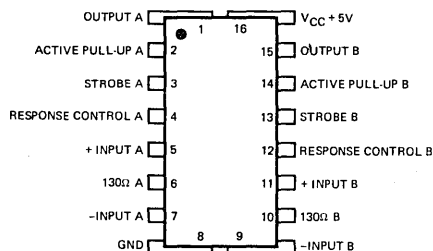
- Dual differential receiver (Am9615) pin-for-pin equivalent to the Fairchild 9615
- Dual differential receiver for single-ended data (Am2615)
- Single 5-volt supply
- High common-mode voltage range (± 15 volts)

- Frequency response control, strobe, and internal terminating resistor
- Am2615 has fail safe capability
- Choice of uncommitted collector or active pull-up outputs
- 100% reliability assurance testing in compliance with MIL-STD-883

The Am2615 is identical to the Am9615 except for the input offset (threshold) voltage. The Am2615 has an input threshold of ~ 1.5 V compatible with DTL & TTL logic. The Am9615 has an input threshold of ~ 0 V. The Am2615 can directly replace the Am9615 and give full safe protection in differential systems where the input difference is > 2.0 V.

The schematic diagram shows a multi-stage electronic circuit. It begins with input terminals 5(11), 6(10), and 8 connected through various resistors (e.g., 1.64k, 8.36k, 7k) to a network of transistors and diodes. A central section includes a bridge-like arrangement of four diodes labeled 3(13) and 4(12). The output stage features two more transistors driving outputs 2(14) and 1(15) through resistors like 80Ω, 4k, and 2.5k. A note at the bottom center specifies "Am2815 ONLY".

Part Number	Package Type	Temperature Range	Order Number
Am2615	Hermetic DIP	-55°C to +125°C	AM2615DM
	Flat Pak	-55°C to +125°C	AM2615FM
	Dice	-55°C to +125°C	AM2615XM
	Hermetic DIP	0°C to +75°C	AM2615DC
	Molded DIP	0°C to +75°C	AM2615PC
	Dice	0°C to +75°C	AM2615XC
Am9615	Hermetic DIP	-55°C to +125°C	9615DM
	Flat Pak	-55°C to +125°C	9615FM
	Dice	-55°C to +125°C	AM9615XM
	Hermetic DIP	0°C to +75°C	9615DC
	Molded DIP	0°C to +75°C	9615PC
	Dice	0°C to +75°C	AM9615XC



NOTE: PIN 1 is marked for orientation.

MAXIMUM RATINGS (Above which the useful life may be impaired)

Storage Temperature	−65°C to +150°C
Temperature (Ambient) Under Bias	−55°C to +125°C
Supply Voltage to Ground Potential (Pin 16 to Pin 8) Continuous	−0.5 V to +7 V
DC Voltage Applied to Outputs for HIGH Output State	−0.5 V to +13.2 V
DC Strobe Input Voltage	−0.5 V to +5.5 V
DC Data Input Voltage	−20 V to +20 V
Output Current, Into Outputs	30 mA
DC Input Current	maximum current is defined by DC Input Voltage

Am2615 ELECTRICAL CHARACTERISTICS OVER OPERATING RANGE

Am2615XM	$V_{CC} = 4.5\text{ V to }5.5\text{ V}$	$T_A = -55^\circ\text{C to }+125^\circ\text{C}$	(MIL grade)
Am2615XC	$V_{CC} = 4.75\text{ V to }5.25\text{ V}$	$T_A = 0^\circ\text{C to }+75^\circ\text{C}$	(COM'L grade)

Parameters	Description	Test Conditions	LIMITS								Units
			$T_A = \text{Min}$		$T_A = 25^\circ\text{C}$			$T_A = \text{Max}$			
			Min	Max	Min	Typ	Max	Min	Max		
V_{OH}	Output HIGH Voltage	$V_{CC} = \text{MIN}, I_{OH} = -5.0\text{ mA}$ $V_{IN+} = +0.8\text{ V}, V_{IN-} = 0\text{ V}$	2.4		2.4	3.2		2.4		Volts	
V_{OL}	Output LOW Voltage	$V_{CC} = \text{MAX}$ $I_{OH} = 15.0\text{ mA}$ $V_{IN+} = +2.0\text{ V}, V_{IN-} = 0\text{ V}$								Volts	
		MIL grade	0.40		.18	0.40		0.40			
		COM'L grade	0.45		.25	0.45		0.45			
I_{CEX}	Output Leakage Current	$V_{CC} = \text{MIN}$ $V_{IN+} = 0\text{ V}$ $V_{IN-} = 4.5\text{ V}$								μA	
		$V_{CEX} = 12\text{ V}$ $V_{CEX} = 5.25\text{ V}$									
		MIL grade									
		COM'L grade	100				100		200		
I_{SC}	Output Short Circuit Current	$V_{CC} = \text{MAX}$ $V_{OUT} = 0\text{ V}$ $V_{IN+} = +0.8\text{ V}$ $V_{IN-} = 0\text{ V}$								mA	
		MIL grade	-15	-80	-15	-39	-80	-15	-80		
		COM'L grade	-14	-100	-14	-39	-100	-14	-100		
I_{IL}	Input Load Current	$V_{CC} = \text{MAX}$ $V_{IN} = V_{OL\text{ MAX}}, \text{ other input} = V_{CC}$		-0.9		-0.49	-0.7		-0.7	mA	
$I_{IL(ST)}$	Strobe Input Low Current	$V_{CC} = \text{MAX}$ $V_{ST} = V_{OL\text{ MAX}}$ $V_{IN+} = +2.0\text{ V}$ $V_{IN-} = 0\text{ V}$		-2.4		-1.15	-2.4		-2.4	mA	
$I_{IL(RC)}$	Response Control Input Load Current	$V_{CC} = \text{MAX}$ $V_{RC} = V_{OL\text{ MAX}}$ $V_{IN+} = +2.0\text{ V}$ $V_{IN-} = 0\text{ V}$				-1.2	-3.4			mA	
V_{CM}	Common Mode Voltage	$V_{CC} = 5.0\text{ V}$ $V_{IN+} - V_{IN-} = 0.4\text{ or }2.4\text{ V}$	-15	+15	-15	± 17.5	+15	-15	+15	V	
$I_{IH(ST)}$	Strobe Input HIGH Current	$V_{CC} = \text{MIN}$ $V_{ST} = 4.5\text{ V}$ $V_{IN+} = +0.8\text{ V}$ $V_{IN-} = 0\text{ V}$								μA	
		MIL grade				2.0		5.0			
		COM'L grade					5.0		10.0		
R_{IN}	Input Resistor	$V_{CC} = 5.0\text{ V}$ $V_{IN+} = 0\text{ V}$ $V_{RES} = 1.0\text{ V}$				77	130	167		Ω	
		MIL grade									
		COM'L grade				74	130	179			
V_{TH}	Differential Input Threshold Voltage	$V_{CM} = 0\text{ V}$	+0.8	+2.0	+0.8	+1.5	+2.0	+0.8	+2.0	V	
I_{CC}	Power Supply Current	$V_{CC} = \text{MAX}$ $V_{IN+} = +2.0\text{ V}$ $V_{IN-} = 0\text{ V}$								mA	
		MIL grade	50		28.7	50		50			
		COM'L grade	50		28.7	50		50			

Switching Characteristics ($T_A = 25^\circ\text{C}$)

Parameters	Test Conditions	Am2615XM			Am2615XC			Units
		Min	Typ	Max	Min	Typ	Max	
t_{pd+} Turn Off Delay	$R_L = 3.9\text{ k}\Omega$	$V_{CC} = 5.0\text{ V}, C_L = 30\text{ pF}$ Refer to figure 4	30	50		30	75	ns
t_{pd-} Turn On Delay	$R_L = 390\text{ }\Omega$		30	50		30	75	
t_{pd+} Turn Off Delay	Strobe to Output	$R_L = 3.9\text{ k}\Omega, C_L = 30\text{ pF}$	7	12		7	15	ns
t_{pd-} Turn On Delay	Strobe to Output	$R_L = 390\text{ }\Omega$	10	15		10	20	

Am2615/9615

Am9615 ELECTRICAL CHARACTERISTICS OVER OPERATING RANGE

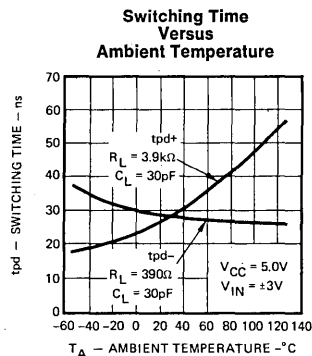
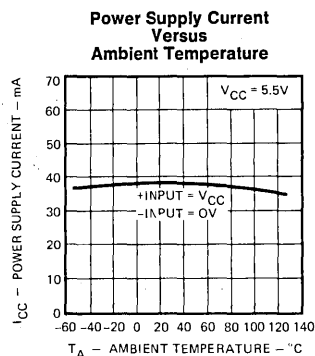
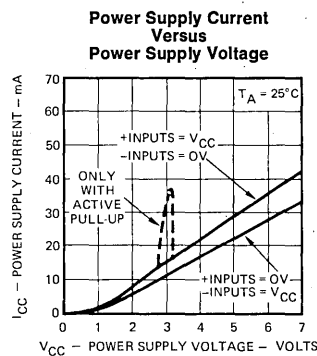
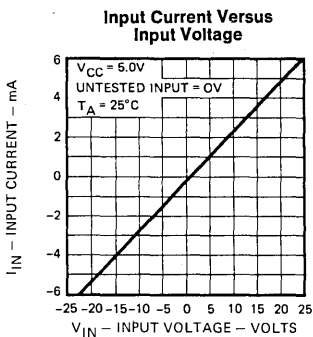
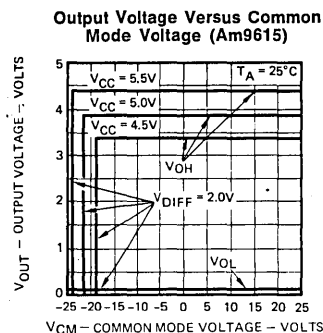
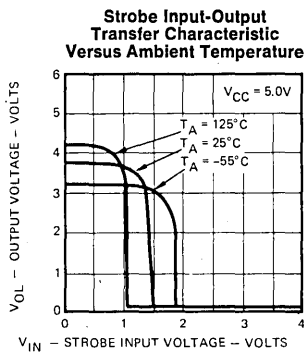
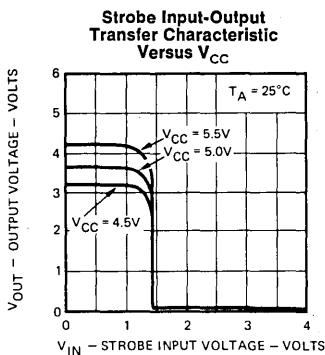
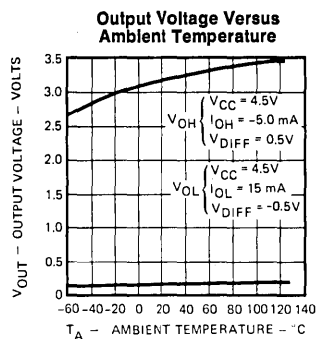
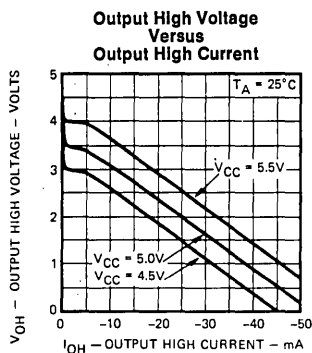
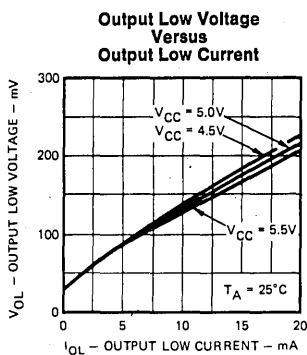
Am9615XM $V_{CC} = 4.5\text{V to } 5.5\text{V}$ $T_A = -55^\circ\text{C to } +125^\circ\text{C}$ (MIL grade)
 Am9615XC $V_{CC} = 4.75\text{V to } 5.25\text{V}$ $T_A = 0^\circ\text{C to } +75^\circ\text{C}$ (COM'L grade)

Parameters	Description	Test Conditions	LIMITS								Units	
			$T_A = \text{Min}$		$T_A = 25^\circ\text{C}$			$T_A = \text{Max}$				
V_{OH}	Output HIGH Voltage	$V_{CC} = \text{MIN}$, $I_{OH} = -5.0 \text{ mA}$ $V_{IN+} = -0.5 \text{ V}$, $V_{IN-} = 0 \text{ V}$	2.4		2.4	3.2		2.4		Volts		
V_{OL}	Output LOW Voltage	$V_{CC} = \text{MAX}$ $I_{OH} = 15.0 \text{ mA}$ $V_{IN+} = +0.5 \text{ V}$, $V_{IN-} = 0$	MIL grade		0.40		.18 0.40	0.40		Volts		
		COM'L grade		0.45		.25 0.45	0.45					
I_{CEX}	* Output Leakage Current	$V_{CC} = \text{MIN}$ $V_{IN+} = 0 \text{ V}$ $V_{IN-} = V_{CC}$	$V_{CEX} = 12 \text{ V}$		MIL grade		100		100		200	μA
		$V_{CEX} = 5.25 \text{ V}$		COM'L grade								
I_{SC}	Output Short Circuit Current	$V_{CC} = \text{MAX}$ $V_{OUT} = 0 \text{ V}$ $V_{IN+} = -0.5 \text{ V}$ $V_{IN-} = 0 \text{ V}$	MIL grade		-15	-80	-15	-39	-80	-15	-80	mA
		COM'L grade		-14	-100	-14	-39	-100	-14	-100		
I_{IL}	Input Load Current	$V_{CC} = \text{MAX}$ $V_{IN} = V_{OL \text{ MAX}}$, other input = V_{CC}			-0.9		-0.49	-0.7		-0.7	mA	
$I_{IL(ST)}$	Strobe Input Low Current	$V_{CC} = \text{MAX}$ $V_{IN+} = +0.5 \text{ V}$ $V_{ST} = V_{OL \text{ MAX}}$ $V_{IN-} = 0 \text{ V}$			-2.4		-1.15	-2.4		-2.4	mA	
$I_{IL(RC)}$	Response Control Input Load Current	$V_{CC} = \text{MAX}$ $V_{IN+} = +0.5 \text{ V}$ $V_{RC} = V_{OL \text{ MAX}}$ $V_{IN-} = 0 \text{ V}$					-1.2	-3.4			mA	
V_{CM}	Common Mode Voltage	$V_{CC} = 5.0 \text{ V}$ $V_{IN+} - V_{IN-} = \pm 2.0 \text{ V}$	-15	+15	-15	± 17.5	+15	-15	+15	V		
$I_{IH(ST)}$	Strobe Input HIGH Current	$V_{CC} = \text{MIN}$ $V_{ST} = 4.5 \text{ V}$ $V_{IN+} = -0.5 \text{ V}$ $V_{IN-} = 0 \text{ V}$	MIL grade					2.0	5.0		μA	
		COM'L grade					5.0	10.0				
R_{IN}	Input Resistor	$V_{CC} = 5.0 \text{ V}$ $V_{IN+} = 0 \text{ V}$ $V_{RES} = 1.0 \text{ V}$	MIL grade		77 130 167					Ω		
		COM'L grade		74 130 179								
V_{TH}	Differential Input Threshold Voltage	$V_{CM} = 0 \text{ V}$	-0.5	+0.5	-0.5	± 0.02	+0.5	-0.5	+0.5	V		
I_{CC}	Power Supply Current	$V_{CC} = \text{MAX}$ $V_{IN+} = +0.5 \text{ V}$ $V_{IN-} = 0 \text{ V}$	MIL grade		50	28.7 50		50		mA		
		COM'L grade		50	28.7 50		50					

Switching Characteristics ($T_A = 25^\circ\text{C}$)

Parameters	Test Conditions	Am9615XM			Am9615XC			Units
		Min	Typ	Max	Min	Typ	Max	
t_{pd+} Turn Off Delay	$R_L = 3.9\text{ k}\Omega$	$V_{CC} = 5.0\text{ V}$, $C_L = 30\text{ pF}$ Refer to figure 4	30	50		30	75	ns
t_{pd-} Turn On Delay	$R_L = 390\text{ }\Omega$		30	50		30	75	
t_{pd+} Turn Off Delay	Strobe to Output	$R_L = 3.9\text{ k}\Omega$, $C_L = 30\text{ pF}$	7	12		7	15	ns
t_{pd-} Turn On Delay	Strobe to Output	$R_L = 390\text{ }\Omega$	10	15		10	20	

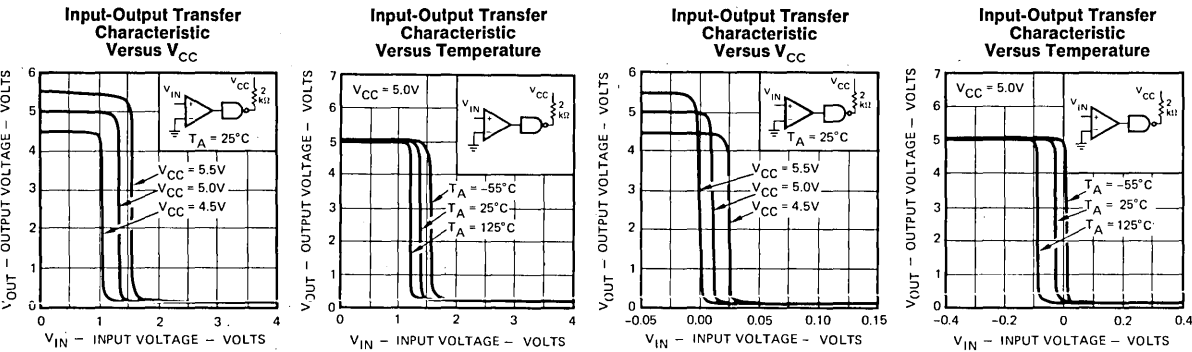
D. C. CHARACTERISTICS



THRESHOLD CHARACTERISTICS

Am2615

Am9615



SWITCHING TIME TEST CIRCUIT & WAVEFORMS

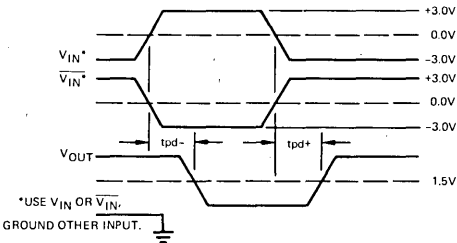
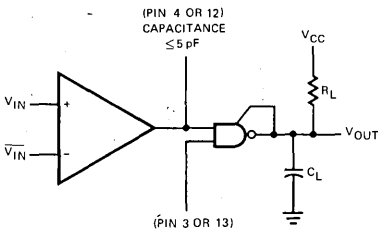
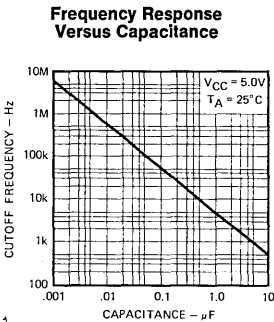
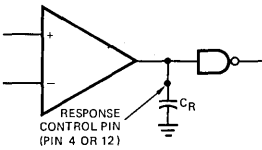


Figure 4

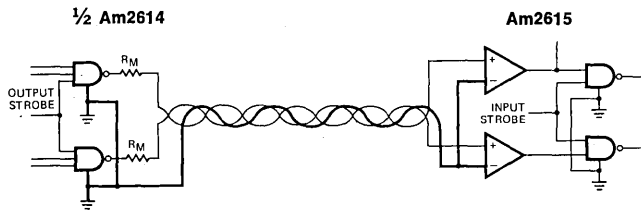
FREQUENCY RESPONSE CONTROL



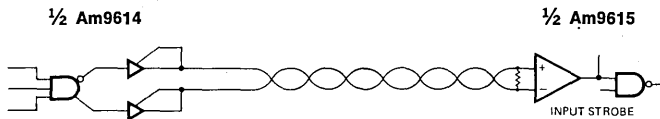
Am2615/9615 LOADING RULES

	Input/Output	Pin No.'s	Input Unit Load	Fanout	
				Output HIGH	Output LOW
Receiver A	Out	1	—	o/c	10
	Active Pull-Up	2	—	83	—
	Response Control	3	—	—	—
	Strobe	4	1.5	—	—
	+ In	5	0.5	—	—
	130 Ω	6	—	—	—
	— In	7	0.5	—	—
Receiver B	GND	8	—	—	—
	— In	9	0.5	—	—
	130 Ω	10	—	—	—
	+ In	11	0.5	—	—
	Response Control	12	—	—	—
	Strobe	13	1.5	—	—
	Active Pull-Up	14	—	83	—
	Out	15	—	o/c	10
	V_{CC}	16	—	—	—

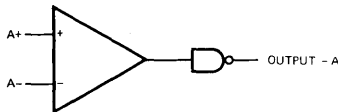
Am2615 STANDARD USAGE
Single-Ended-Back Matched Operation
With Common Ground



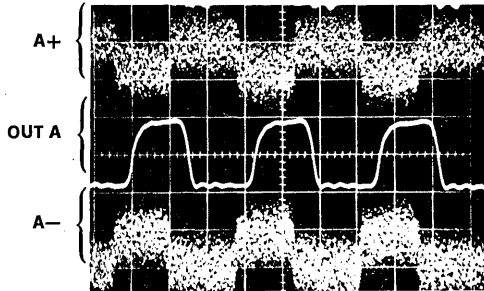
Am9615 STANDARD USAGE
Differential Operation



Photograph of an Am9615 switching differential data in the presence of high common mode noise.

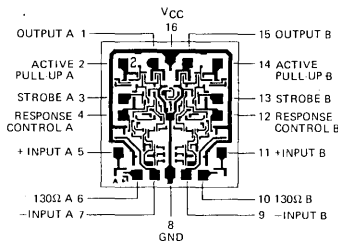


Vertical = 2.0 V/Div. Horizontal = 50 ns/Div.



4

Metallization and Pad Layout



53 X 58 Mils